

NIMBUS-9³/₄

HARDWARE FULL-STACK — DRAWING PACKAGE

Real Quidditch · broom + three balls · BOM · OpenSCAD CAD · drawings

SHEET 1	BROOM	manned eVTOL, 8 internal ducted fans
SHEET 2	GOLDEN SNITCH	evasion drone, real flapping wings
SHEET 3	BLUDGER	pursuit + soft-tag, zero contact
SHEET 4	QUAFFLE	catch / carry / throw / score
SHEET 5	MATCH FLEET	rollup cost for one 7v7 match

"Magic? No. We just wrote the collision-avoidance really well."

NIMBUS-9¾ BROOM

Manned eVTOL — 1 rider, 8 ducted fans, fly-by-intent

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Overall length2400 mm

core/params.py LEN = 2.4 m

Lift fans8 × EDF (internal)

_default_fan_layout (exact x/y)

Thrust-to-weight2.45 (hover @ 41%)

analysis/flight_thermal.py

Battery sprint~61 s / charge

2600 Wh alone — a sprinter

Match endurance~26 min (hybrid)

series-hybrid: analysis/endurance_match.py

Cooling margin8.1× waste heat

lift air = coolant + heat pipes

League ceiling150 m (svc ~km)

altitude_ceiling.py + fence_ceiling

Control inputwave-to-fly (gesture)

firmware/rider_input.py -> intent mapper

NOTE

Disc area is capped by what fits inside the bristle flare (the fan never protrudes), so battery-only it's a ~60 s sprinter. A slim in-shaft SERIES-HYBRID genset burning SAF lifts that to a ~26 min match (the battery becomes the peak buffer, full ~250 kW bursts kept). Fans

internal, lift air = coolant. And you fly it by WAVING the handle, pose -> RiderIntent -> the existing intent mapper. FLIGHT + THERMAL + MATCH all PASS.

BILL OF MATERIALS — BRM (41 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
BRM-01	Carbon-fibre monocoque body p...	2.4 m, hides the fan bay; SI->mm x1000 in cad/broom.scad	1	2,200	2,200
BRM-02	Handle shaft + brass pommel	tapered Ø34->Ø22 mm, leather-wrapped grip	1	180	180
BRM-03	Bristle fairing leaf	~780 carbon-straw strands, aerodynamic shroud	1	640	640
BRM-04	Electric ducted fan (EDF) unit	360 N max each; main-lift in bristle shroud + in-shaft trim fans	8	520	4,160
BRM-05	Brushless motor for EDF	~12 kW peak, 6S-14S, duct-matched	8	340	2,720
BRM-06	ESC (electronic speed control...	200 A, telemetry, 1st-order ~40 ms response	8	150	1,200
BRM-07	Li-ion buffer battery pack	2600 Wh, serves the ~250 kW peaks in the hybrid	1	3,100	3,100
BRM-08	Battery management system (BM...	14S smart BMS, cell balancing + SoC telemetry	1	260	260
BRM-09	Power distribution board	800 A bus, redundant rails	1	180	180
BRM-10	Flight controller MCU	Cortex-M4F (thumbv7em-none-eabihf), 400 Hz loop	1	95	95
BRM-11	IMU (redundant)	triple-redundant 6-DOF, 1 kHz	3	55	165
BRM-12	Magnetometer	3-axis, maneuver-compensated	2	18	36
BRM-13	Barometer	high-res, for altitude hold	2	12	24
BRM-14	RTK-GPS receiver + antenna	cm-grade, dual-band	1	420	420
BRM-15	UWB ranging tag	indoor cm positioning to pitch anchors	2	40	80
BRM-16	Solid-state LiDAR	360deg, edge collision avoidance	2	600	1,200
BRM-17	Low-latency datalink radio	redundant 5G + dedicated mesh	2	140	280
BRM-18	Ballistic recovery parachute ...	Ø7.6 m canopy, rocket-deployed, full-altitude envelope	1	2,200	2,200
BRM-19	Airbag skirt	underbody impact airbag, impact-triggered	1	450	450
BRM-20	Master kill-switch receiver	referee e-stop -> gentle descent	1	60	60
BRM-21	5-point harness + saddle	straddle seat, energy-absorbing	1	320	320
BRM-22	Rider intent controls	handlebar grips, throttle, lean sensors	1	210	210
BRM-23	Brass compass/attitude gauge	cosmetic + real attitude readout	1	90	90
BRM-24	LED trajectory strip	addressable, night-match light show	1	70	70
BRM-25	Heat pipe (sintered copper)	Ø6 mm, full-shaft length, motor->shaft spreader	4	22	88
BRM-26	Air-over-motor duct shroud	routes lift airflow across windings (air = coolant)	8	16	128
BRM-27	In-duct winding temp sensor	NTC, one per fan, feeds the governor	8	3	24
BRM-28	Thermal interface material	phase-change pad, motor + ESC to spreader	1	18	18
BRM-29	DShot ESC wiring harness	bidirectional DShot600, 8 channels + telem	1	45	45
BRM-30	CAN-FD vehicle bus	5 Mbit, BMS + ESC telem + pose to Dementor	1	30	30
BRM-31	Hardware abstraction layer	firmware skin around the Rust core	1	0	0
BRM-32	Micro-turbine genset	~125 kW continuous, slim in-shaft series-hybrid range extender	1	9,800	9,800
BRM-33	FADEC + start system	full-authority turbine control + spin-up	1	1,200	1,200
BRM-34	SAF fuel bladder + lines	~18 L crash-safe bladder inside the shaft	1	340	340
BRM-35	Sustainable aviation fuel	15 kg SAF per fill (~48x li-ion energy density)	1	30	30
BRM-36	Exhaust ejector + heat exchan...	entrains bypass air, drops exhaust <150 C	1	420	420
BRM-37	Ceramic-matrix bristle tips	high-temp bristles at the exhaust path	1	260	260
BRM-38	Gesture input transducer	handle IMU + twist grip -> RiderIntent	1	0	0
BRM-39	Air-data system	pitot + density/OAT probe for altitude compensation	1	180	180
BRM-40	Reserve parachute	second canopy, independent deploy	1	1,400	1,400
BRM-41	High-ceiling league profile	fence_ceiling = 150 m (vs 18 m demo default)	1	0	0

UNIT TOTAL (prototype):

\$34,303

DRAWING

NIMBUS-9¾

2026-06-27

UNITS

mm / SI

SCALE

NTS

PART

BRM

REV

v0.1

CAD

cad/broom.scad

GOLDEN SNITCH

Walnut-sized evasion drone — beamed-power, full-match flight

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius40 mm (Ø80)

ball/params.py snitch.radius = 0.04 m

Mass target50 g (kept)

snitch.mass = 0.05 kg

Top speed19 m/s

snitch.max_speed

Max accel32 m/s²

snitch.max_accel (absurdly agile)

Lift4 × micro-EDF (internal)

swarm core – the real flight

Battery sprint~2.7 min

1S 300 mAh alone – ≈ the fatigue_tau

Match endurancecontinuous (beamed)

analysis/snitch_endurance.py

Pitch power~2.5 kW array

5.8 GHz steered to the broadcast pose

NOTE

WINGS ARE THEATRE (SNT-02/03/04 flutter; lift is the internal EDF swarm SNT-05 — don't tell Potterheads). A 50 g walnut can't carry fuel like the broom, so it HARVESTS energy instead: the gold dimple-vents double as a 5.8 GHz rectenna (SNT-15/16) and a pitch array beams power onto its broadcast pose (SNT-19/20). The cell becomes a buffer (jukes on a supercap SNT-17). Form + agility kept, match filled, no pit.

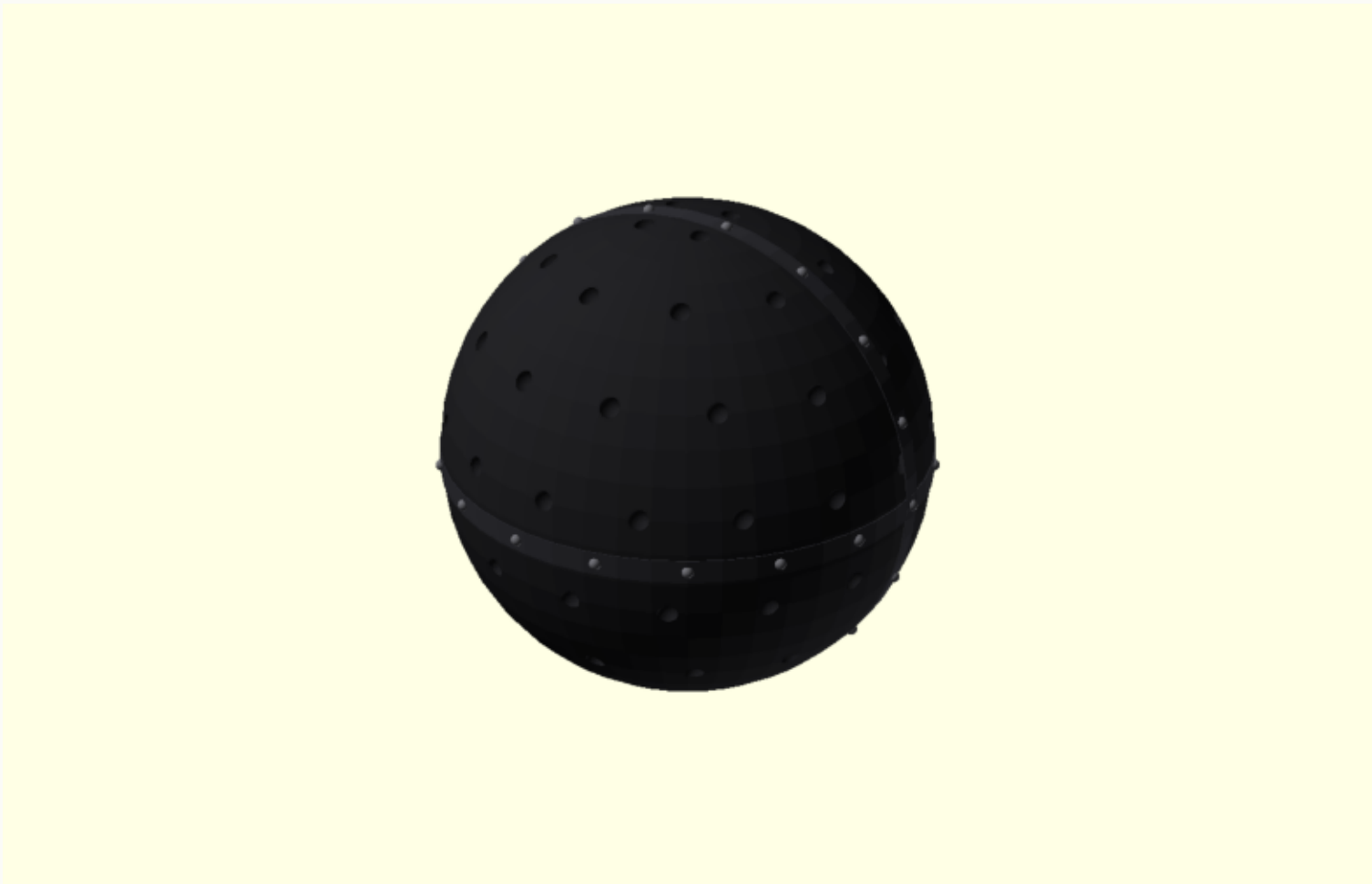
BILL OF MATERIALS — SNT (20 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
SNT-01	Gold-clad vented sphere shell	Ø80 mm (R=40 mm), walnut-sized, dimple vents	1	90	90
SNT-02	Animatronic flapping wing ass...	thin filigree vane on hinge, span ~76 mm	2	28	56
SNT-03	Coreless gearmotor + crank	6 mm coreless, flap linkage, 8-14 Hz beat	2	9	18
SNT-04	Wing membrane	5 micron gold-tint mylar, feather-cut	2	4	8
SNT-05	Micro ducted-fan lift unit	Ø16 mm internal, the ACTUAL lift	4	22	88
SNT-06	Micro brushless motor	4 mm, paired to micro-EDF	4	11	44
SNT-07	Micro ESC (4-in-1)	5 A, integrated	1	26	26
SNT-08	Nano flight controller	STM32 class, IMU on-board, predictive evasion	1	38	38
SNT-09	Optical-flow + ToF sensor	360deg-ish reach detection	2	16	32
SNT-10	Micro LiDAR / proximity ring	detects incoming hands/airframes	1	45	45
SNT-11	Capacitive capture sensor	registers an enclosing hand	1	12	12
SNT-12	Micro Li-Po cell	1S 300 mAh, now a ride-through BUFFER (not the engine)	1	8	8
SNT-13	Gold micro-LED	glow so the crowd can track it	1	5	5
SNT-14	2.4 GHz telemetry link	reports pose to the Dementor referee	1	14	14
SNT-15	Conformal rectenna array	5.8 GHz, printed into the gold dimple-vent pattern	1	34	34
SNT-16	RF->DC harvester PMIC + MPPT	rectifier + maximum-power tracking, 5.8 GHz	1	17	17
SNT-17	Graphene supercapacitor	7 F @ 5.4 V, ~1-2 g — the PEAK buffer	1	12	12
SNT-18	In-shell power-bus + balancing	rectenna / supercap / cell tri-source manager	1	9	9
SNT-19	Beam-steer retro-beacon	lets the pitch array lock its spot onto the Snitch	1	11	11
SNT-20	Pitch power-beaming array (pe...	5.8 GHz phased array, ~6 m^2 aperture, ~2.5 kW wall	1	0	0

UNIT TOTAL (prototype):

\$567

ISOMETRIC



SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius150 mm (Ø300)

ball/params.py bludger.radius = 0.15 m

Mass1.2 kg (foam)

bludger.mass – looks iron, isn't

Top speed15 m/s

bludger.max_speed

Max accel22 m/s²

bludger.max_accel (most aggressive)

Lift4 × EDF (internal)

swarm core

Soft-tag radius1.5 m

tag_radius (outside no-contact floor)

Bat deflect1.6 m

bat_radius

Contact0 — software hit

ball/avoidance.py, worst clr +0.40 m

NOTE

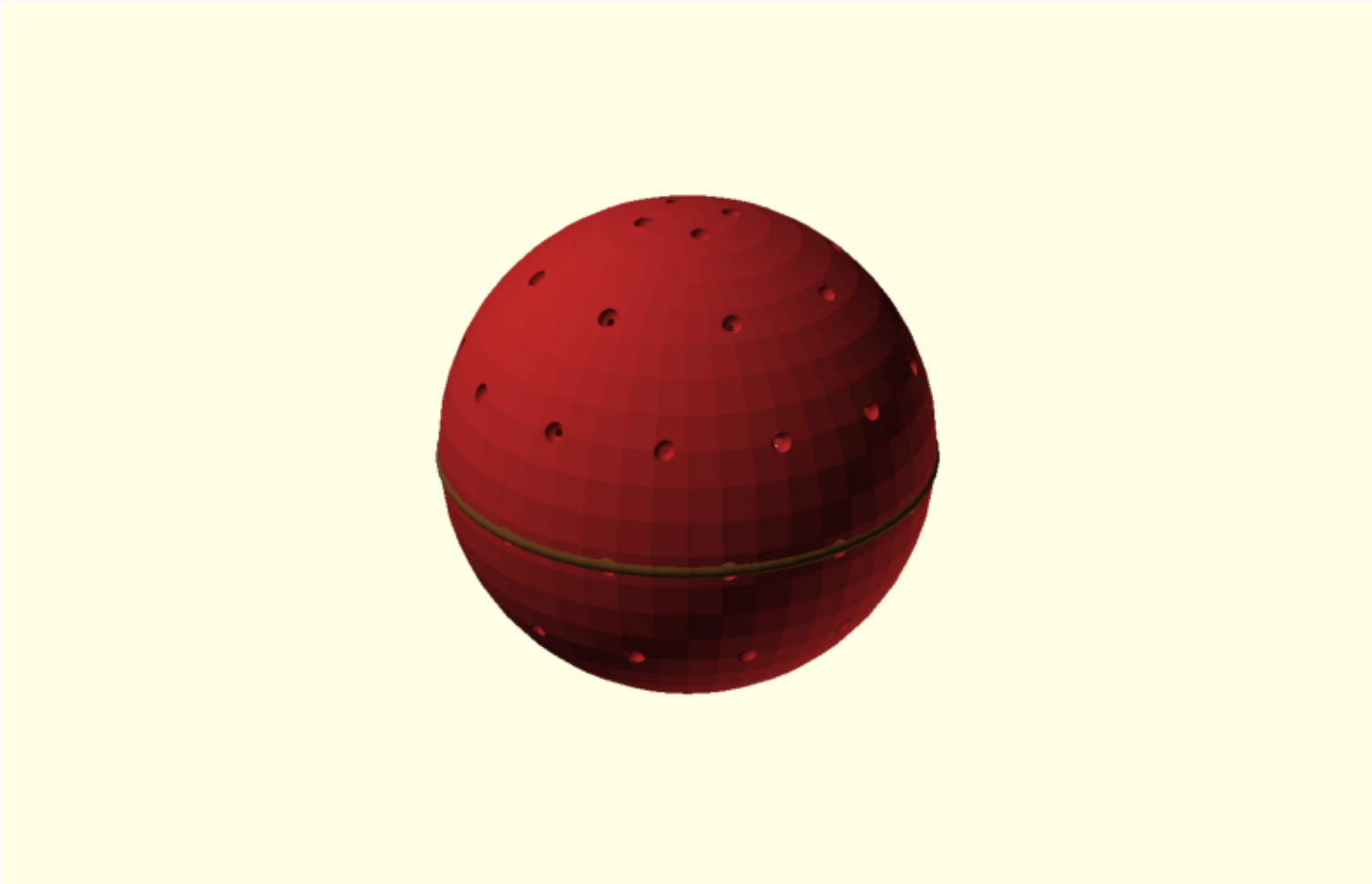
All threat, no contact. The 'hit' is a proximity tag registered in software ~1.5 m out; the foam shell never has to reach a person.

BILL OF MATERIALS — BLG (12 line items, prototype qty 1)

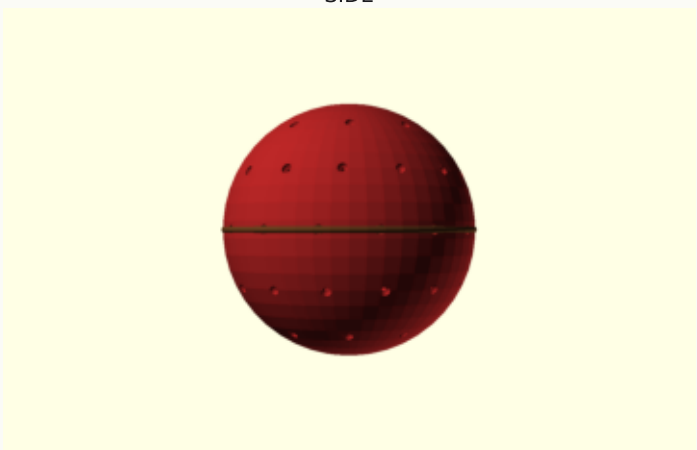
REF	PART	SPEC	QTY	\$ EA	\$ EXT
BLG-01	Black impact-foam sphere shell	Ø300 mm (R=150 mm), riveted-iron look	1	70	70
BLG-02	Equator rivet bands (cosmetic)	two crossed bands + dummy rivets	1	18	18
BLG-03	Ducted-fan lift unit	Ø40 mm internal, internal swarm	4	40	160
BLG-04	Brushless motor	per fan, high power-to-weight	4	22	88
BLG-05	4-in-1 ESC	20 A, telemetry	1	38	38
BLG-06	Flight controller	IMU on-board, lead-pursuit guidance	1	42	42
BLG-07	LiDAR / proximity sensor	target ranging for the soft tag	2	55	110
BLG-08	Bat-swing detector	sees the beater's marked bat	1	16	16
BLG-09	No-contact avoidance compute	clamps velocity to a stop at the shell	1	0	0
BLG-10	Li-Po pack	3S, sized for sprint pursuits	1	24	24
BLG-11	Status LED	angry red glow	1	5	5
BLG-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype):\$585

ISOMETRIC



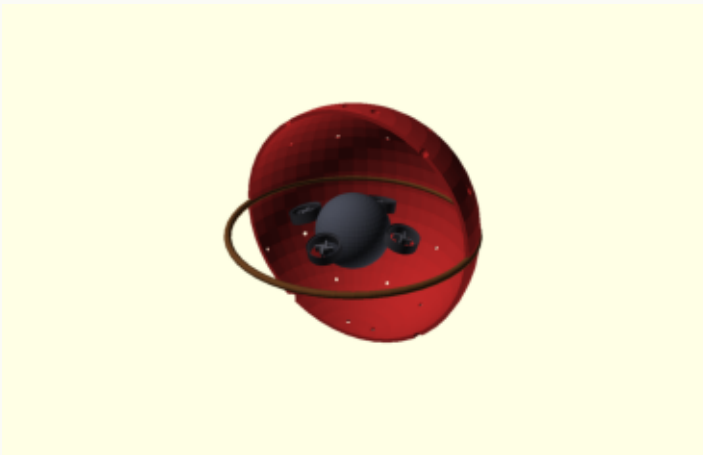
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius180 mm (Ø360)

ball/params.py quaffle.radius = 0.18 m

Mass0.5 kg

quaffle.mass – light despite its size

Top speed7 m/s

quaffle.max_speed (gentle)

Catch radius0.45 m

quaffle.catch_radius (forgiving)

Lift4 × EDF (internal)

swarm core, soft hover

On catchthrust → 0 in hand

ball/quaffle.py capacitive sense

Scoringhoop optical gate

auto pass-through detection

Safetyno-contact floor

even the nice ball won't bonk you

NOTE

Soft hover, generous catch, rides with its holder, then throws to the nearest hoop with tunable aim-assist (on for youth, off for pros).

BILL OF MATERIALS — QAF (12 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
QAF-01	Red grippy foam sphere shell	Ø360 mm (R=180 mm), seamed + finger grooves	1	75	75
QAF-02	Grab-handle ring	recessed leather ring, holds the catch sensor	1	14	14
QAF-03	Ducted-fan lift unit	Ø48 mm internal, internal swarm	4	42	168
QAF-04	Brushless motor	low-speed, smooth	4	20	80
QAF-05	4-in-1 ESC	15 A, telemetry	1	34	34
QAF-06	Flight controller	IMU on-board, throw + aim-assist	1	40	40
QAF-07	Capacitive catch sensor array	detects a holding hand -> thrust to 0	1	18	18
QAF-08	Hoop pass-through beacon	IR tag read by the hoop optical gate	1	10	10
QAF-09	No-contact avoidance compute	even the nice ball won't bonk you	1	0	0
QAF-10	Li-Po pack	3S, long hover endurance	1	22	22
QAF-11	Status LED	warm white possession glow	1	5	5
QAF-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype):\$480

ITEM	CODE	MATCH QTY	UNIT \$	EXT \$
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NIMBUS-9¾ BROOM	BRM	14	\$34,303	\$480,242
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QUAFFLE	QAF	1	\$480	\$480
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BLUDGER	BLG	2	\$585	\$1,170
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GOLDEN SNITCH	SNT	1	\$567	\$567
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MATCH TOTAL			\$482,459
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